Aedin Lai

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Education

Northeastern University Expected May 2029

Candidate for Bachelor of Science in Electrical & Computer Engineering — Boston, MA
Coursework: Calculus **2**, Physics **2**, Chemistry, Cornerstone of Engineering **2**, First-Year Writing

Experience

SingleStore June – September 2026

Data Center Infrastructure Intern — Boston, MA
- Led hardware deployment for a phased data center expansion, scaling a private cloud platform from **160** to **300+** VMs
- Redesigned a single-push rollout into an incremental "two racks at a time" strategy to cut disruption to live workloads
- Delivered capacity across three cohorts of **40–50** VMs, hitting or landing within one week of every target date
- Diagnosed a Linux kernel storage defect and a multi-week NFS instability, deploying fixes that restored full fleet capacity and system stability
- Supported a full Oakland-to-Boston data center relocation, migrating the platform from bare-metal to virtualized infrastructure

PEACE Foundation, Yale University School of Medicine 2023 – 2024

Research Intern — New Haven, CT
Supervisor: Dr. Kun Liu, Associate Professor of Sports Neuroscience
- Designed, built, and improved a full-body balance measurement device to study cognition and movement in neurological patients
- Automated data collection using custom hardware across ASD, Alzheimer's, Parkinson's, and schizophrenia patient populations
- Presented findings at ISEC **2024** and co-authored results accepted for publication in JAMA **(2025)**
- Co-developed a Yale study assessing stressors and emotional health in Division I, III, and club collegiate fencing athletes

Projects

One-Inch Punch Arcade Machine Fall 2025

Lead Constructor & Designer — Northeastern University, GE1501
- Worked hands-on for the whole build — mechanical construction, electronics, and scoring software — from concept to expo launch
- Designed a "one-inch punch" mechanism using dual laser sensors, and wrote the scoring logic converting swing velocity into a live score
- Built the wooden frame, pendulum-and-magnet reset system, and laser-cut acrylic paneling entirely from scratch
- Validated durability through **80+** personal stress-test swings, catching and fixing structural wobble before launch
- Built a layered-sponge padding system that safely absorbed harder hits than originally scoped
- Identified a paddle material limit from full-force hits during a full expo day of play and proposed a rubber-composite redesign

Cal Tracker September 2025

Self-hosted calorie tracker — Python, HTML/CSS, Raspberry Pi
- Built and self-hosted a real-time calorie tracking web app on a Raspberry Pi with a lightweight front-end/back-end and persistent local storage

Hallway Marathon Event Site November 2025

hallway-marathon.github.io
- Designed and built an event website from scratch (HTML, CSS, JavaScript, no frameworks) with live betting odds and competitor stats for a multi-university competition
- Independently managed all content, deployment, and live updates throughout the event

Leadership & athletics

Northeastern University Club Fencing 2025 – Present

Captain
- Anchor the epee team to the **2026** Collegiate Fencing Club National Championship
- Lead team operations — scheduling, logistics, recruitment, and practice design — for the full roster
- Serve as primary contact for tournament registration, travel, and university representation at competitions

Marx Fencing Academy 2018 – 2025

Competitive epee fencer
- USA Fencing rating **A25**; ranked **#103** nationally and **#122** globally
- Top-**16** World Cup finish; All-American **2021–2025**

Skills

Programming	Python, Java, C#, SQL, MATLAB, HTML/CSS, JavaScript
Hardware & CAD	Raspberry Pi, embedded systems, sensor integration, SolidWorks, AutoCAD
Tools	Git, GitHub, Pandas, Linux, VS Code

Live projects: aedinlai.com/gatecraft, /sandbit, /hand-playground, /doodle-life, /games, /gym, /broadcast, and football.aedinlai.com